Problem 2.1 (a)

$$H = \frac{\vec{p}^{\,2}}{2m} + \frac{1}{2} k \vec{x}^{\,2}$$

$$Z_N = \frac{1}{N!} \int \prod_{j=1}^{N} \frac{d\vec{p}_j \, d\vec{x}_j}{(2\pi\hbar)^3} e^{-\beta H} = \frac{1}{N!} (Z_1)^N$$

where one-particle partition function is

$$Z_1 = \int \frac{d\vec{p}}{(2\pi\hbar)^3} \int d\vec{x} \; e^{-\beta\left(\frac{\vec{p}^{\,2}}{2m} + \frac{1}{2}k\vec{x}^{\,2}\right)} =$$

$$= \int \frac{dp_x \, dp_y \, dp_z}{(2\pi\hbar)^3} e^{-\frac{\beta}{2m}(p_x^2 + p_y^2 + p_z^2)} \int dx\,dy\,dz \; e^{-\frac{1}{2}\beta k (x^2+y^2+z^2)}$$

$$= \left(\int_{-\infty}^{\infty} \frac{dp_x}{2\pi\hbar} e^{-\frac{\beta p_x^2}{2m}} \right)^3 \left(\int_{-\infty}^{\infty} dx \; e^{-\frac{1}{2}\beta k x^2} \right)^3 =$$

$$p_x = \sqrt{\frac{2m}{\beta}}\, q \qquad x = \sqrt{\frac{2}{\beta k}}\, u$$

$$= \left(\frac{1}{2\pi\hbar} \sqrt{\frac{4m}{k\beta^2}} \int_{-\infty}^{\infty} dq \, e^{-q^2} \int_{-\infty}^{\infty} du \, e^{-u^2} \right)^3 = \left(\frac{2\pi k_B T}{2\pi\hbar\omega_0} \right)^3 =$$

$$k = m\omega_0^2 \Rightarrow \sqrt{\frac{4m}{k}} = \frac{2}{\omega_0} \qquad = (\hbar\omega_0\beta)^{-3}.$$

Average energy per particle $\frac{E}{N} = -\frac{\frac{\partial}{\partial\beta} \ln Z}{N} = -\frac{\partial}{\partial\beta} \frac{1}{N} \ln \frac{(Z_1)^N}{N!} =$

$$= -\frac{\partial}{\partial\beta} \ln Z_1 = \frac{\partial}{\partial\beta} \ln (\hbar\omega_0\beta)^3 = \frac{3}{\beta} = 3k_B T.$$

Therefore heat capacity per oscillator is $C_1 = \frac{\partial}{\partial T} E_1 = 3k_B$.

For N of them, $C_V = 3k_B N$.

Prob. 2.2

(a) Just follow textbook, chap. 2.2.2 and 2.2.3.

$$\langle E \rangle = 3 \frac{L^3}{(2\pi)^3} \int d^3\vec{k}\ \hbar\omega(\vec{k}) \left(n_B(\omega(\vec{k})) + \frac{1}{2} \right)$$

3 independent polarizations

$$d^3\vec{k} = 4\pi k^2 dk$$

$$\omega(\vec{k}) = v k$$

$$= 3 \frac{4\pi L^3}{(2\pi)^3 v^3} \int_0^\infty d\omega\ \omega^2 \hbar\omega \left(n_B(\omega) + \frac{1}{2} \right) = \int_0^\infty d\omega\ g(\omega)\ \hbar\omega \left(n_B(\omega) + \frac{1}{2} \right).$$

n = atom density, $nL^3 = N$ = total number of atoms.

$$g(\omega) = N \frac{9\omega^2}{\omega_d^3}, \quad \omega_d = \left(6\pi^2 n v^3\right)^{1/3},$$

Unrestricted integral over ω will result in $\langle E \rangle \sim T^4$ (and $C_V \sim T^3$) at all temperatures.

Debye: need to cutoff the integral

$$\Rightarrow \langle E \rangle = \int_0^{\omega_d} d\omega\ g(\omega)\ \hbar\omega \left(n_B(\omega) + \frac{1}{2} \right)$$

$$\hookrightarrow \frac{1}{e^{\hbar\omega\beta} - 1}$$

The last term (from $\frac{1}{2}$ factor under the integral is T-independent

$\Rightarrow$ we can drop it.

$$\langle E \rangle = \frac{9N\hbar}{\omega_d^3} \int_0^{\omega_d} d\omega\ \omega^2\ \omega\ \frac{1}{e^{\beta\hbar\omega} - 1} = \frac{9N\hbar}{\omega_d^3} (\hbar\beta)^{-4} \int_0^{\hbar\omega_d\beta} dx\ \frac{x^3}{e^x - 1}$$

Let $x = \hbar\omega\beta \Rightarrow \omega = \frac{x}{\hbar\beta}$

$$= 9N \frac{(k_B T)^4}{(\hbar\omega_d)^3} \cdot \int_0^{\frac{\hbar\omega_d}{k_B T}} dx\ \frac{x^3}{e^x - 1}$$

Low-temperature: $\hbar\omega_d \gg k_B T \Rightarrow$ upper limit of integration can be replaced by ∞.

$$J = \int_0^\infty \frac{dx\, x^3}{e^x - 1} = \int dx\, x^3 \frac{e^{-x}}{1-e^{-x}} = \int_0^\infty dx\, x^3 e^{-x} \sum_{n=0}^{\infty} e^{-nx}$$

$$= \sum_{n=1}^{\infty} \int_0^\infty dx\, x^3 e^{-nx} = \sum_{n=1}^{\infty} \frac{1}{n^4} \underbrace{\int_0^\infty dy\, y^3 e^{-y}}_{6} =$$

let $x = \frac{y}{n}$

$$= 6 \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{15}.$$

Thus $\langle E \rangle = \frac{9N}{\hbar^3 \omega_d^3} (k_B T)^4 \frac{\pi^4}{15}$

$$C_V = \frac{9N}{\hbar^3 \omega_d^3} \frac{4\pi^4}{15} (k_B T)^3 k_B = N k_B \left(\frac{T}{T_d}\right)^3 \frac{12\pi^4}{5}.$$

(b) Plot $C(T)$ data from the table as C vs. T^3!

The data points form a very good straight line.

Linear fit gives

$$C = \underbrace{0.00092}_{N k_B \frac{12\pi^4}{5} \frac{1}{T_d^3}} \cdot T^3$$

gas constant $= 8.31 \frac{J}{K\,mol}$

$$T_d^3 = 8.31 \cdot \frac{12\pi^4}{5} \cdot \frac{1}{0.00092} \approx 2.11 \cdot 10^6$$

$\Rightarrow T_d \approx 128$ K. This agrees nicely with

$T_d = 129 \pm 2$ K quoted in Phys. Rev. vol. 112 p. 45 (1958)

Debye theory works well here.

Prob. 2.3 Two-dimensional solid

We assume that the solid, which is really like a piece of graphene, is sitting on the top of some massive 3d material. The van der Waals interaction between 2d and 3d materials strongly confines atoms of our 2d solid to move only in a two-dimensional plane. The energy cost of them moving out of plane is very high. Under this assumption (which is opposite to the assumption of free-standing film) there are only 2 polarizations for atomic oscillations.

(<u>Note</u>: it is okay to make a different assumption, the difference will affect only numerical coefficients.)

Then

$$\langle E \rangle = 2 \sum_{\vec{k}} \hbar\omega(\vec{k}) \left(n_B(\omega(\vec{k})) + \frac{1}{2} \right)$$

sum over 2d wave vectors only, $\vec{k} = (k_x, k_y, 0)$

$$= 2 \frac{L^2}{(2\pi)^2} \int d^2\vec{k}\, \hbar\omega(\vec{k}) \left(n_B(\omega(\vec{k})) + \cancel{\frac{1}{2}} \right)$$

we can drop this T-independent piece right away.

$$\omega(\vec{k}) = vk \qquad d^2\vec{k} = 2\pi k\, dk = \frac{2\pi}{v^2}\omega\, d\omega$$

$$\langle E \rangle = 2 \frac{2\pi L^2}{(2\pi)^2 v^2} \int_0^{\omega_d} d\omega\, \omega (\hbar\omega) \frac{1}{e^{\beta\hbar\omega} - 1}$$

We integrate to ω_d which we determine by requiring that at high temperature $k_B T \gg \hbar\omega_d$, the energy must be $\langle E \rangle = 2 k_B T N$ (1 k_B per coordinate per particle)

<u>At high-T</u> $e^{\beta\hbar\omega} = e^{\frac{\hbar\omega}{k_B T}} \approx 1 + \frac{\hbar\omega}{k_B T} \Rightarrow n_B(\omega) \to \frac{k_B T}{\hbar\omega}$

Then $\langle E \rangle = \frac{4\pi L^2}{(2\pi)^2 v^2} \int_0^{\omega_d} d\omega\, \omega\, \hbar\omega \frac{k_B T}{\hbar\omega} =$

$$= \frac{L^2}{\pi v^2} k_B T \int_0^{\omega_d} d\omega\, \omega = \underbrace{\frac{L^2 \omega_d^2}{2\pi v^2}}_{2N} k_B T \Rightarrow \omega_d^2 = \frac{4\pi v^2 N}{L^2}$$

$$= 4\pi v^2 n_2$$

$n_2 = \frac{N}{L^2}$ = two-dimensional density

Now we study <u>low-T</u> $(k_B T \ll \hbar\omega_d)$

$$\langle E \rangle = \frac{4\pi L^2}{(2\pi)^2 v^2} \int_0^{\omega_d} d\omega \frac{\hbar\omega^2}{e^{\beta\hbar\omega}-1} = \frac{L^2 \hbar}{\pi v^2} \left(\frac{k_B T}{\hbar}\right)^3 \int_0^{x_d} \frac{dx\, x^2}{e^x - 1}$$

$$\beta\hbar\omega = x, \quad \omega = \frac{x}{\hbar\beta} = \frac{k_B T}{\hbar} x$$

$$x_d = \beta\hbar\omega_d = \frac{\hbar\omega_d}{k_B T} \gg 1$$

So we take $x_d \to \infty$ and use $\frac{L^2}{\pi v^2} = \frac{4N}{\omega_d^2}$

$$\Rightarrow \langle E \rangle = 4N \frac{(k_B T)^3}{(\hbar\omega_d)^2} \cdot K, \quad K = \int_0^\infty dx \frac{x^2}{e^x - 1} = 2\zeta(3) \cong 2{,}4$$

$$C = \frac{\partial E}{\partial T} = 12 N K \left(\frac{k_B T}{\hbar\omega_d}\right)^2.$$

Importantly, $C \sim T^2$ as a result of two-dimensional nature of the material.

<u>Summary</u>: $d=3$ solid $C \sim T^3$

$d=2$ $C \sim T^2$

$d=1$ $C \sim T$ (not calculated here but easy to show)